

Shokunin Technical Overview

v4.2.2

github.com/EliasOulkadi/shokunin

62 Skills | Windows + Linux | Open Source MIT

Architecture Overview

Shokunin is a production-grade AI coding ecosystem providing persistent memory via ChromaDB, 62 specialized AI skills, MCP (Model Context Protocol) servers for tool integration, and a declarative self-update system. It runs fully offline with zero server infrastructure and zero API costs.

Memory System

The memory system uses a three-layer architecture: ChromaDB for vector semantic search across sessions, JSONL transaction logs for structured programmatic analysis, and Markdown session files for human-readable review. The MCP server implements JSON-RPC 2.0 over stdin/stdout with 9 tools.

- `store_context`: Store text in persistent memory with type, tags, and project metadata
- `search_context`: Vector similarity search with freshness decay blending
- `get_session_summary`: Retrieve all entries for a specific session (max 100)
- `multi_search_context`: Combined vector + BM25 + temporal filter with RRF fusion
- `consolidate_memories`: Summarize old entries per project
- `list_sessions`: Recent sessions with metadata and entry counts
- `continue_session`: Full context load for session resume
- `save_message`: Record messages in session transcript
- `verify_file_path`: Validate file/directory existence before acting on stale claims

Freshness Decay

Memories decay exponentially over a 30-day half-life. The `search_context` and `multi_search_context` tools accept a `freshness_boost` parameter (0.0 to 1.0) that blends vector similarity with recency. At 0.0, results are purely relevance-based. At 1.0, they are purely recency-based. At 0.5, the blend gives equal weight to relevance and freshness. This prevents stale memories from drowning out recent context.

Claim Verification

The hardest problem in persistent memory is not storage -- it is retrieval shaping. When an agent recalls 'the auth helper lives at src/x.ts' from a memory written two weeks ago, it cannot act blindly. The `verify_file_path` MCP tool validates file paths before the agent proceeds. If the file no longer exists, the agent searches memory for newer entries describing the refactoring, then greps the actual codebase. Memory is treated as a claim from a frozen point in time -- never as fact.

Skills Ecosystem -- 62 Skills

Every skill includes YAML frontmatter with name and description for automatic trigger matching. All 62 skills now include four mandatory sections: Workflow (numbered procedural steps), Error Handling (5+ Cause|Fix scenarios), Sources (5+ verifiable references), and Anti-Patterns (5+ Pattern|Problem|Fix entries). Average skill length is 257 lines.

- Infrastructure: docker, kubernetes, terraform, ci-cd, db-admin
- Backend: api-forge, auth-architect, db-sculptor, error-handler
- Frontend: component-forge, responsive-engine, motion-craft, landing-craft, aesthetic-web, ui-ux-pro-max, email-design-eng, impeccable, taste, taste-soft, taste-minimalist
- Mobile: flutter, react-native
- Quality: test-commander, performance-profiler, code-review, comprehensive-review, cross-review, zen-review, zen-comprehensive-review
- Content: communication, content-marketing, business-proposals, seo-geo, translate-craft, documentation
- Productivity: git-workflow, windows-powershell, runbook-gen, strategy, design, finance, legal-counsel, whendone-plus
- Documents: kami, kagen, portfolio-auto
- AI Agents: agent-browser, agent-tools, skill-creator
- Extras: playwright, web-security, plan, find-skills, efficient-coding, senior-engineer
- Ecosystem: shokunin-update, chromadb, memory
- Language: humanize, research, init

Getting Started

Windows one-command install:

```
irm https://raw.githubusercontent.com/EliasOulkadi/shokunin/master/install.ps1 | iex
```

Linux:

```
bash <(curl -sSL https://raw.githubusercontent.com/EliasOulkadi/shokunin/master/install.sh)
```

Requirements: Windows 10/11 or Linux, Node.js 18+, Python 3.11+, Git. After installation, open a new terminal and run 'opencode'. The ecosystem activates automatically.

Quality and Testing

- 29 integration tests covering security (path traversal sanitization), MCP protocol compliance, and ChromaDB operations (save/search/recall/session)

- 12 PowerShell scripts with Set-StrictMode, ErrorActionPreference, and CmdletBinding
- 100% type-annotated Python with lazy ChromaDB initialization
- CI/CD pipeline with mypy type checking, bandit security scanning, and dependabot
- Cross-platform support: Windows 11 (native) and Linux (via install.sh)